

Initial conditions for a superposition of sound waves

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1 The equations of motion

Let's use the equations (11) and (12) from [2], where we neglect of the scalar field terms :

$$\dot{E} + \partial_i(EV^i) + p \left[\dot{W} + \partial_i(WV^i) \right] = 0 \quad (1)$$

$$\dot{Z}_i + \partial_j(Z_i V^j) + \partial_i p = 0 \quad (2)$$

where $E = W\epsilon$, $Z_i = W^2(\epsilon + p)V_i$ and W is the relativistic Lorentz factor. Let's make the following assumptions :

- $W = 1$ everywhere, we are in the non-relativistic limit
- $3p = \epsilon$, the fluid is a perfect relativistic fluid
- the velocity is a sound wave

At the beginning of the simulation, the velocity field is initialized taking into account its power spectrum. However this is not enough to specify that the velocity field will evolve as the superimposition of soundwave. The energy density has to be initialized correctly to allow the velocity field to evolve as the superimposition of soundwaves.

The abovementionned equation (2) gives

$$V_i \dot{\epsilon} + \epsilon \dot{V}_i + V_i \partial_j(\epsilon V^j) + \epsilon V_j \partial_j(V_i) + \frac{1}{4} \partial_i \epsilon = 0 \quad (3)$$

while the equation (1) gives

$$\dot{\epsilon} + \partial_j(\epsilon V^j) + \frac{1}{3} \epsilon \partial_j(V^j) = 0 \quad (4)$$

Combining the two allows us to write the evolution equation for the velocity field

$$\boxed{\dot{V}_i - \frac{1}{3} V_i \partial_j(V^j) + V_j \partial_j(V_i) + \frac{1}{4} \frac{\partial_i \epsilon}{\epsilon} = 0} \quad (5)$$

2 What was done before

In earlier calculations, we made the assumption that both the velocity field and the energy density were sound waves, i.e. that the energy density was of the form

$$\epsilon = \epsilon_0 + \sum_{\vec{k}} \epsilon_k e^{i\vec{k} \cdot \vec{x}} \quad (6)$$

In the lowest order, this allows to link the velocity power spectrum with the energy density power spectrum and initialize the simulation. However this leads to dramatic results when the velocities are high. Indeed, for high velocities the perturbation becomes bigger than the mean value ϵ_0 and the energy density becomes negative leading to a crash of the simulation.

3 Proper initial conditions for sound waves

The structure of equation 5 suggests to introduce an intermediate field λ such that

$$\lambda = \ln \left(\frac{\epsilon}{\epsilon_0} \right) = \sum_k \lambda_k e^{ik \cdot x} \quad (7)$$

Then this equation can be rewritten as

$$\dot{V}_i - \frac{1}{3} V_i \partial_j (V^j) + V_j \partial_j (V_i) + \frac{1}{4} \partial_i (\lambda) = 0 \quad (8)$$

neglecting the non-linear term for now, one can relate $\dot{V}_k$ and λ_k

$$\dot{V}_i + i \frac{1}{4} k^i \lambda_k = 0 \quad (9)$$

Contracting with k_i

$$\lambda_k = \frac{4i}{k^2} k_i \dot{V}_i \quad (10)$$

Our input is the power spectrum of the velocity V_k which is linked to the power spectrum of its derivative, though the two are independent random variables

$$\omega \frac{d \langle V_k^2 \rangle}{d \ln k} = \frac{d \langle \dot{V}_k^2 \rangle}{d \ln k} \quad (11)$$

$$\boxed{\epsilon = \epsilon_0 \prod_{\vec{k}} \exp(\lambda_k e^{ik \cdot x}) > 0} \quad (12)$$

This ensures, at least in the linear order, that the velocity field is the sum of independent sound waves with positive energy. One can check that for zero velocity, the energy density is equally ϵ_0 .

4 Initial conditions for a purely vortical motion

The fundamental equation is the following

$$\dot{V}_i - \frac{1}{3} V_i \partial_j (V^j) + V_j \partial_j (V_i) + \frac{1}{4} \partial_i (\lambda) = 0 \quad (13)$$

At the linear order we want to suppress the sound waves, leading to

$$\Delta \lambda = 0 \quad (14)$$

5 Comment on lattice momenta

For a good discussion on how to define derivatives, discrete Fourier transforms and lattice momenta, see [1] especially section 2. Based on this material, it is clear that instead of k^i , one should use k_{eff}^i defined equation (2.15)

$$k_{eff}^i = 2 \frac{\sin(\pi \tilde{n}_i / N)}{\delta_x} \quad (15)$$

However in the simulation a different definition is used

$$k_{eff}^i = \sqrt{\frac{2 - 2 \cos(2\pi \tilde{n}_i / N)}{\delta_x^2}} \quad (16)$$

Thankfully the two are equal ¹ but it seems to me that the second accumulates more numerical error. Shouldn't we use the first one?

¹Using $\cos(2x) = 1 - 2 \sin^2(x)$

References

- [1] Daniel G. Figueroa, Juan Garcia-Bellido and Arttu Rajantie. On the Transverse-Traceless Projection in Lattice Simulations of Gravitational Wave Production. *Journal of Cosmology and Astroparticle Physics*, 2011(11):pp. 015–015, 2011. ISSN 1475-7516. doi:10.1088/1475-7516/2011/11/015. 1110.0337.
- [2] Mark Hindmarsh, Stephan J. Huber, Kari Rummukainen and David J. Weir. Numerical simulations of acoustically generated gravitational waves at a first order phase transition. *Physical Review D - Particles, Fields, Gravitation and Cosmology*, 92(12), 2015. doi:10.1103/PhysRevD.92.123009.